

Active Directory Lab: Account Administration and Security Policy

Lab Project by Tyrese Johnson

Overview

Active Directory (AD) is Microsoft's centralized directory service to manage users, computers, and services within a Windows domain environment. Many corporations and educational institutions use AD for identity management, authentication, and authorization. The purpose of this lab is to demonstrate helpdesk and network administration skills, as well as device-hardening cybersecurity skills to prevent unauthorized access generally and improper use by users in the domain.

Devices	Relevant IP Address
Windows Server 2022 – Active Directory Domain Controller	172.16.0.1
Windows 11 Computer (sometimes referred to as client machine)	172.16.0.100

Domain Controller: I bypass the set-up process for the Active Directory in this lab report, but all anyone would need to do is go into the Server Manager, navigate to the Tools in the upper right corner, and download Active Directory Domain Services. After that, you would need to update the computer to your Domain Controller. As shown below, I titled my computer “**MYSERVER**” and the Active Directory domain itself “**MyHomeLab.local**.”

Server Manager

Server Manager ▸ AD DS

Dashboard
Local Server
All Servers
AD DS
DNS
File and Storage Services ▸

SERVICES
All servers | 1 total

Filter

Server Name	IPv4 Address	Manageability	Last Update	Windows Activation
MYSERVER	10.0.2.15, 172.16.0.1	Online - Performance counters not started	10/2/2026 10:01:54 AM	00454-40000-00001-AA620 (Act...

EVENTS
All events | 6 total

Filter

Server Name	ID	Severity	Source	Log	Date and Time
MYSERVER	4013	Warning	Microsoft-Windows-DNS-Server-Service	DNS Server	10/2/2026 9:56:18 AM
MYSERVER	2886	Warning	Microsoft-Windows-ActiveDirectory_DomainService	Directory Service	10/2/2026 9:56:09 AM
MYSERVER	3041	Warning	Microsoft-Windows-ActiveDirectory_DomainService	Directory Service	10/2/2026 9:56:09 AM
MYSERVER	3054	Warning	Microsoft-Windows-ActiveDirectory_DomainService	Directory Service	10/2/2026 9:55:57 AM
MYSERVER	3051	Warning	Microsoft-Windows-ActiveDirectory_DomainService	Directory Service	10/2/2026 9:55:57 AM
MYSERVER	1539	Warning	Microsoft-Windows-ActiveDirectory_DomainService	Directory Service	10/2/2026 9:55:56 AM

Type here to search

10:04 AM
10/2/2026

Recycle Bin

Active Directory Users and Computers

File Action View Help

Active Directory Users and Com
Saved Queries
MyHomeLab.local
Built-in
Company Machines
Computers
Domain Controllers
Employees
ForeignSecurityPrincipal...
Managed Service Account...
Users

Name	Type	Description
Builtin	builtinDomain	
Company M...	Organizational...	
Computers	Container	Default container for up...
Domain Con...	Organizational...	Default container for do...
Employees	Organizational...	
ForeignSecu...	Container	Default container for sec...
Managed Se...	Container	Default container for ma...
Users	Container	Default container for up...

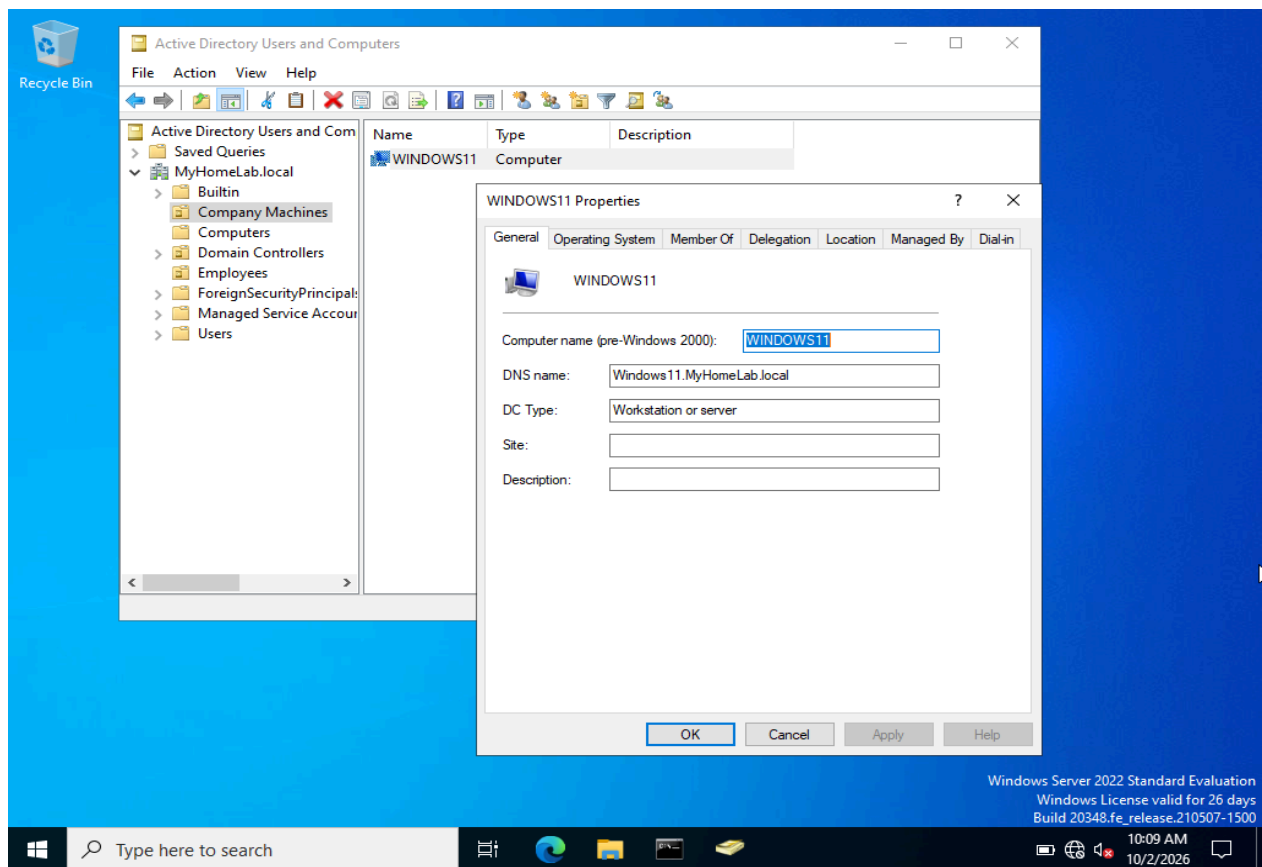
Windows Server 2022 Standard Evaluation
Windows License valid for 26 days
Build 20348.fe_release.210507-1500

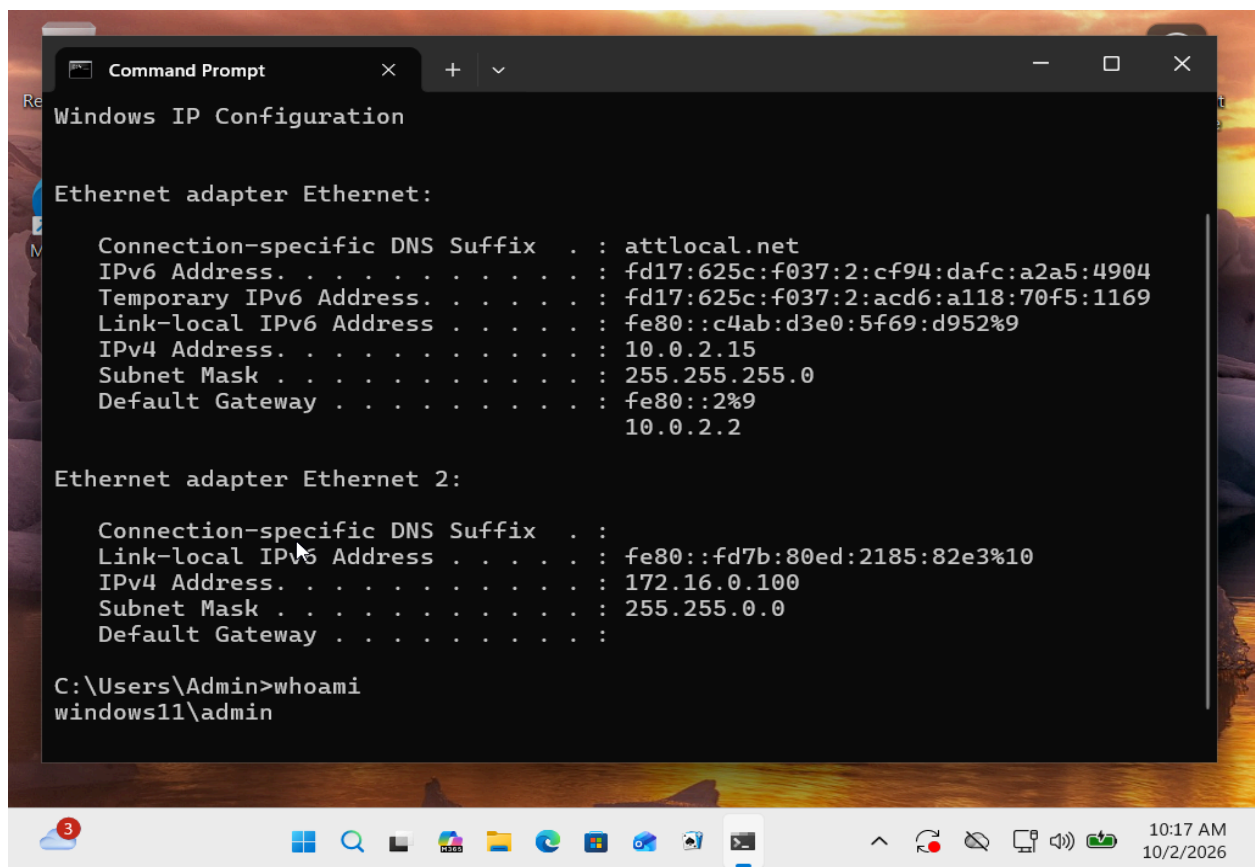
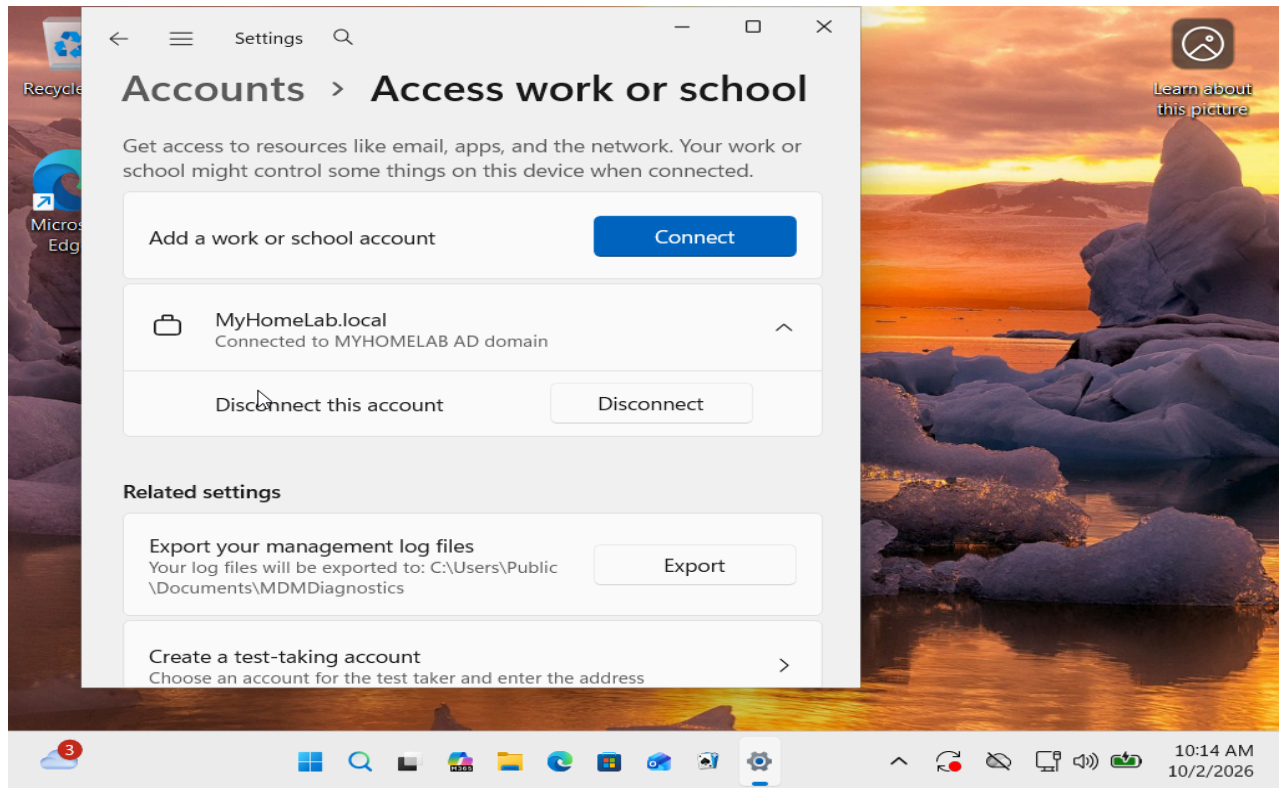
Type here to search

10:06 AM
10/2/2026

Instead of placing my computers in the “Computer” folder, I decided to create an Organizational Unit (OU) called “Company Machines.” Similarly, I did not place my users in the “Users” folder, but the “Employees” OU.

The below shows my Windows 11 computer in Company Machines from the perspectives of both the domain controller and the client machine joined to the domain. A third screenshot shows the Windows 11 IP address from the command prompt.

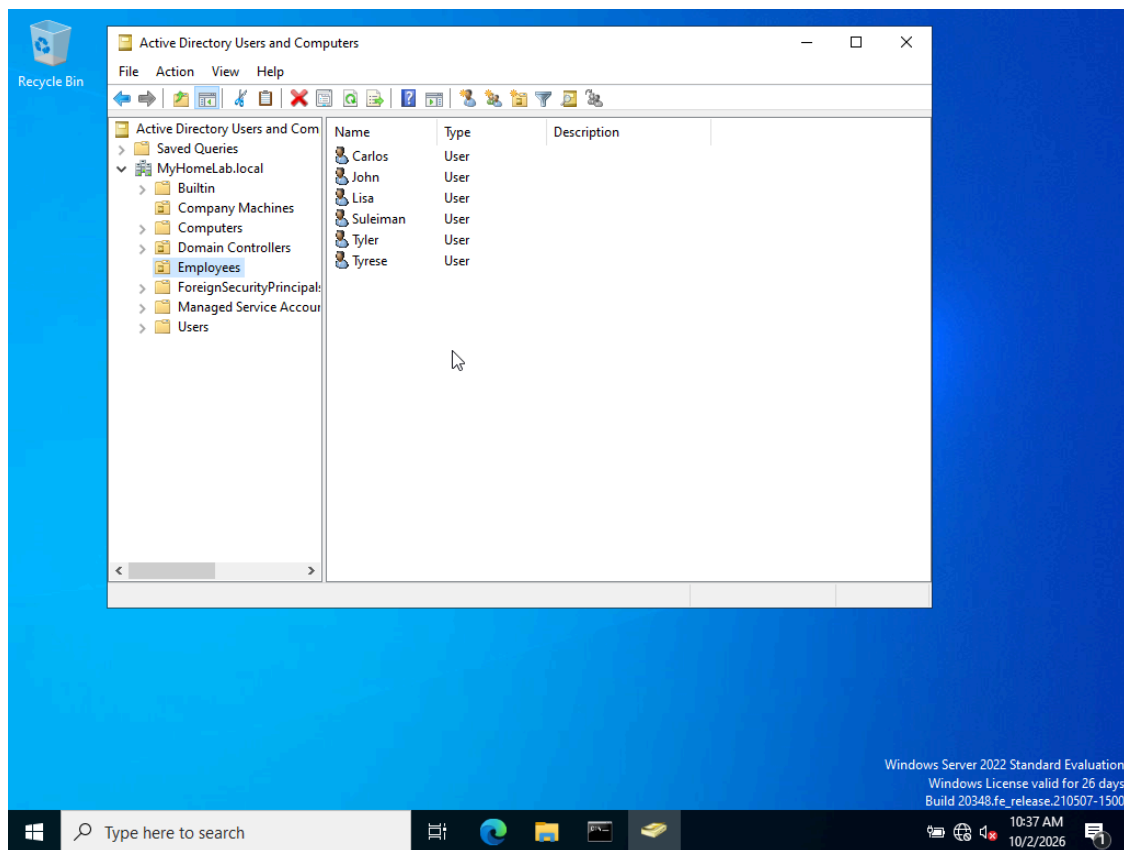




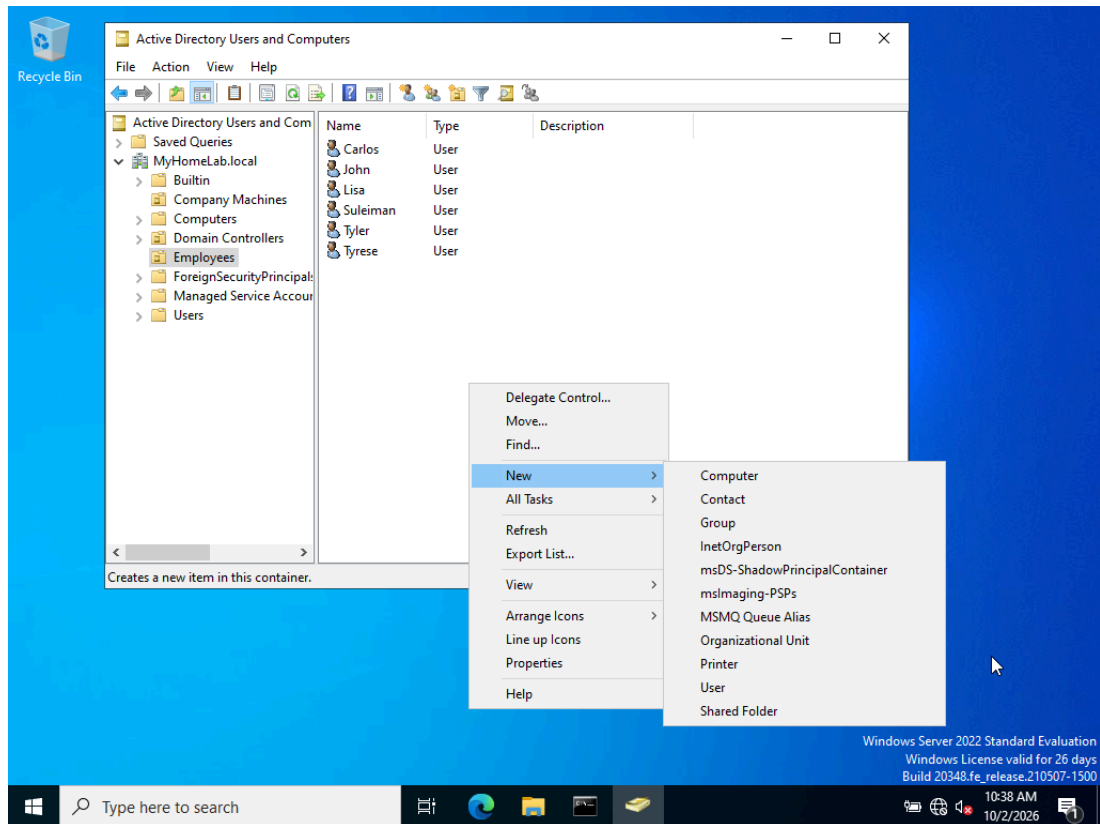
Employees OU

I created six employees, which includes their usernames and passwords.

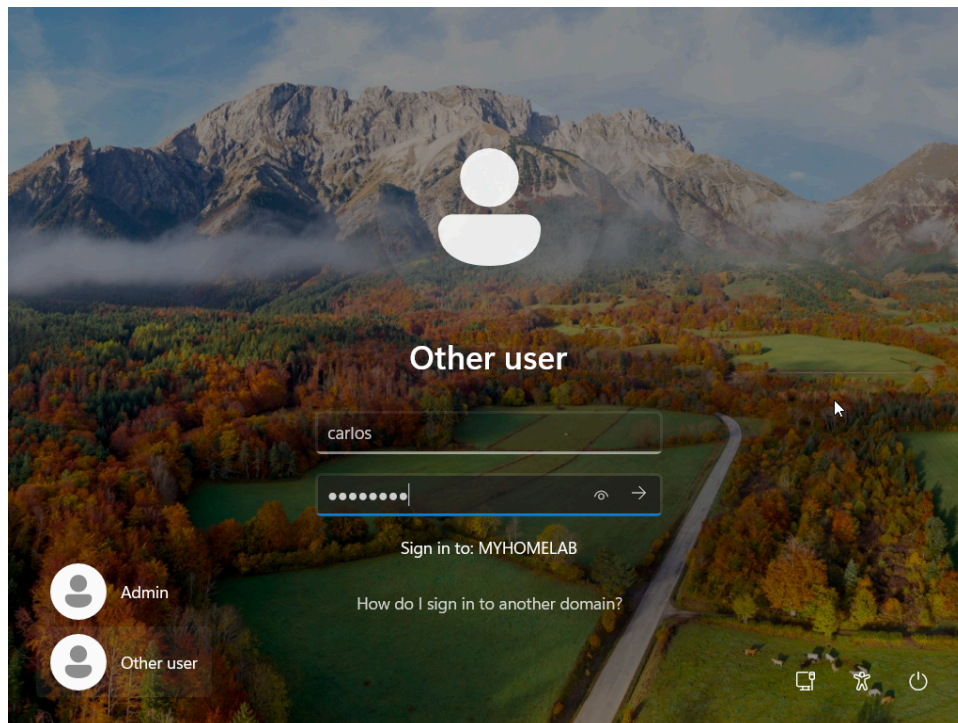
Carlos	Password: [redacted]
John	Password: [redacted]
Lisa	Password: [redacted]
Suleiman	Password: [redacted]
Tyler	Password: [redacted]
Tyrese	Password: [redacted]



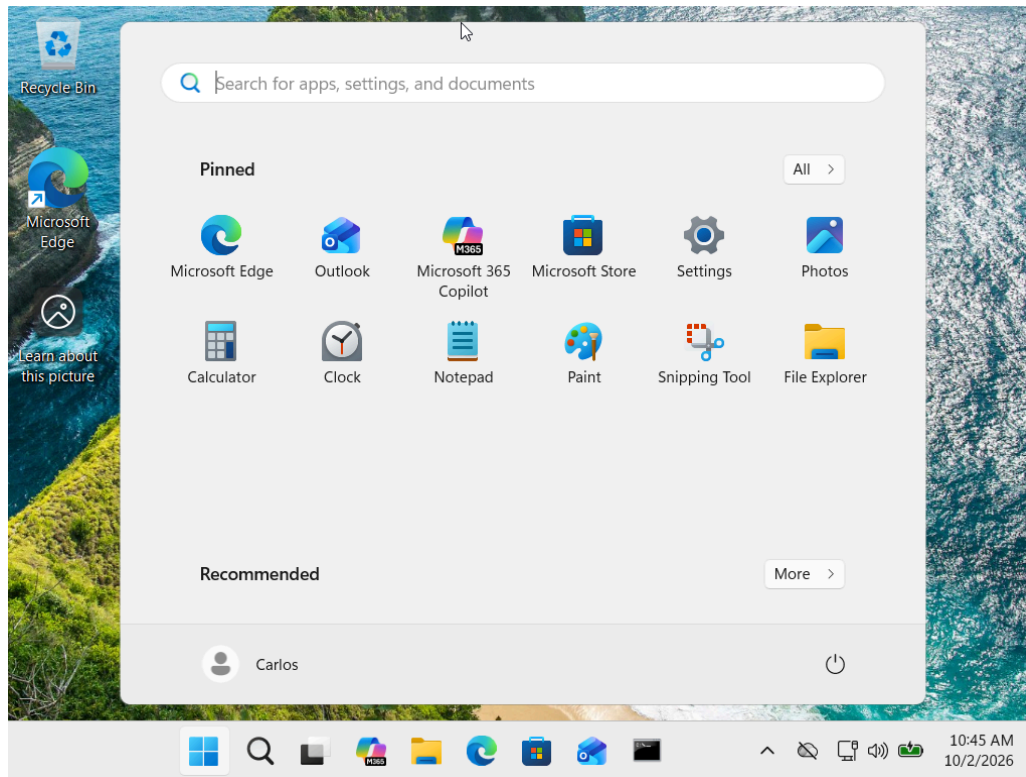
Just right-click to create a new user, or anything else. (Shown below.)



Sign-in to the MyHomeLab domain on the Company Machine:



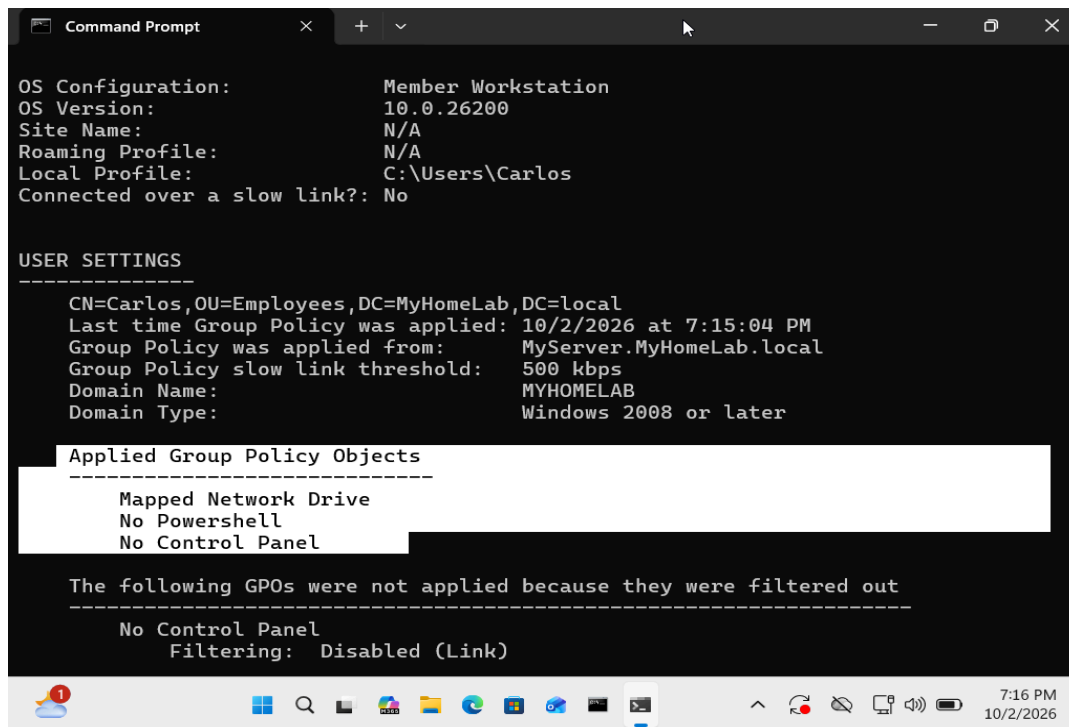
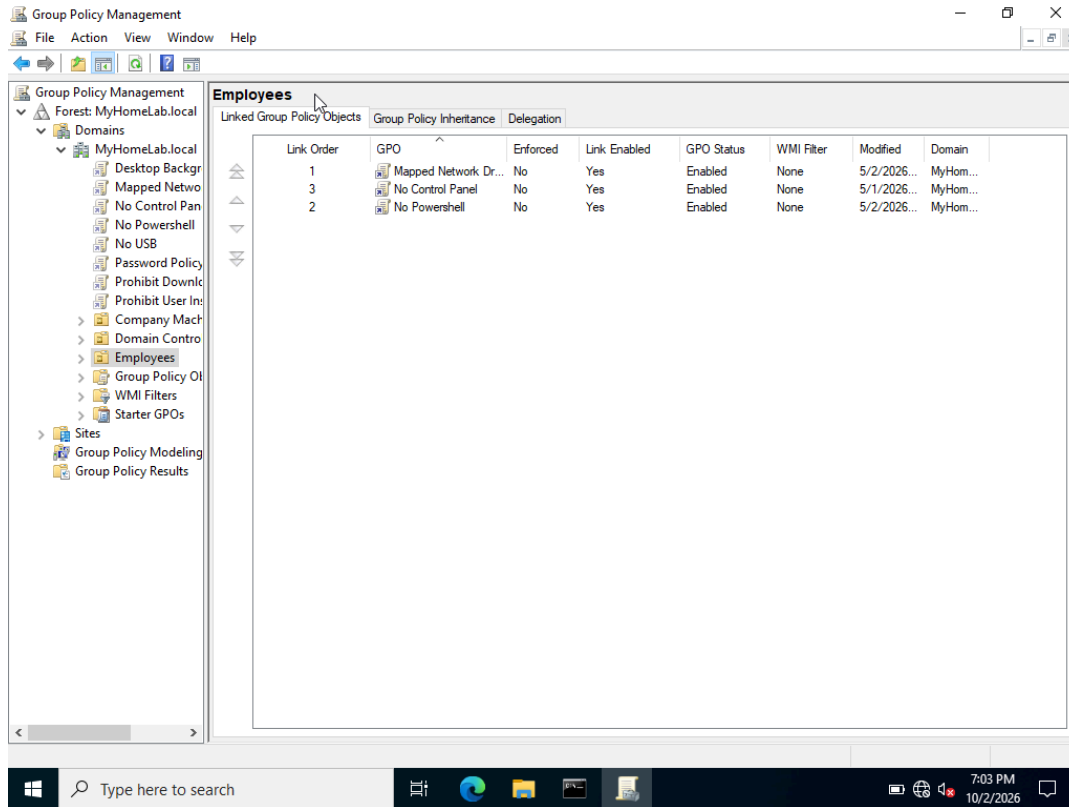
Successful login



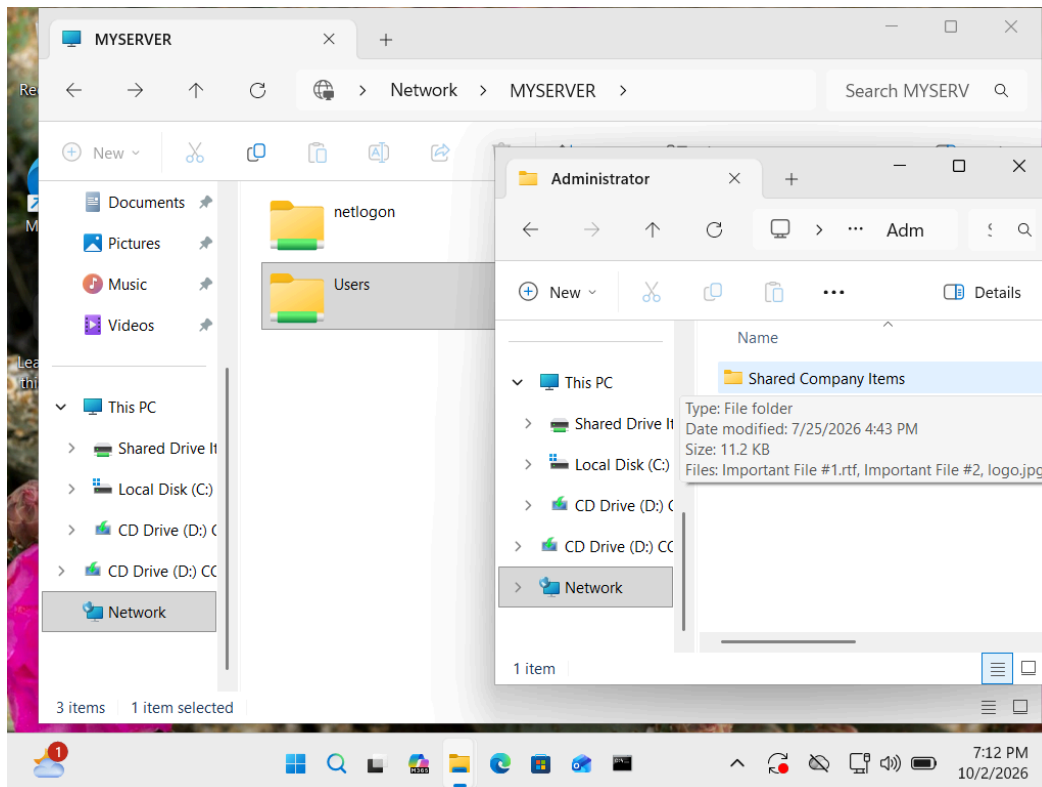
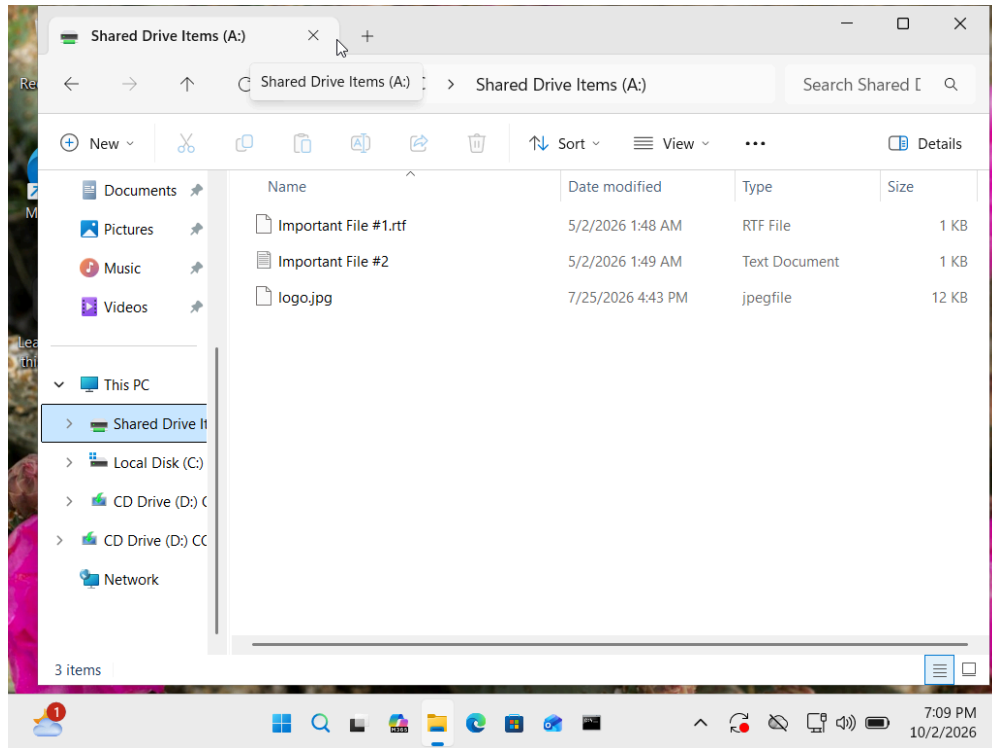
Group Policies and Demonstrations: Some group policies are applied to users specifically, and others to computers. We will start with the users-only group policies.

GPO 1: Shared Network Drive	Gives all users on the network access to a shared folder.
GPO 2: No Control Panel	Blocks access to the Control Panel for all users
GPO 3: No Powershell	Blocks access to the Powershell application for all users

Below shows the user group policies from the domain controller/server, as well as the client machine.

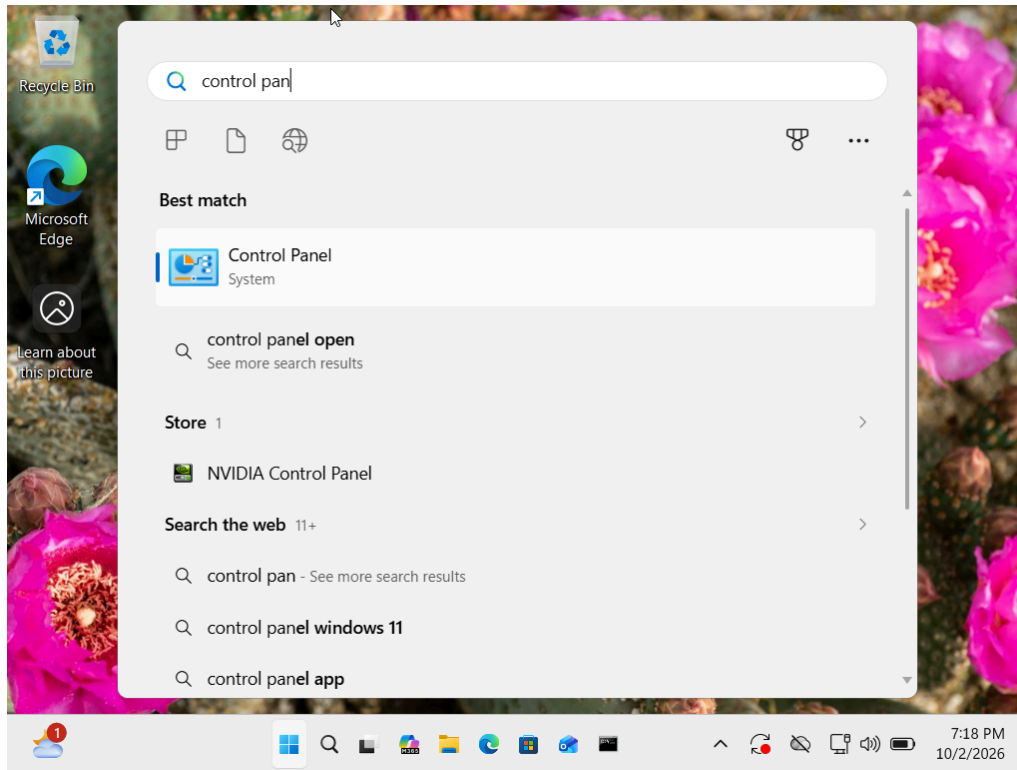


GPO 1: Shared Network Drive

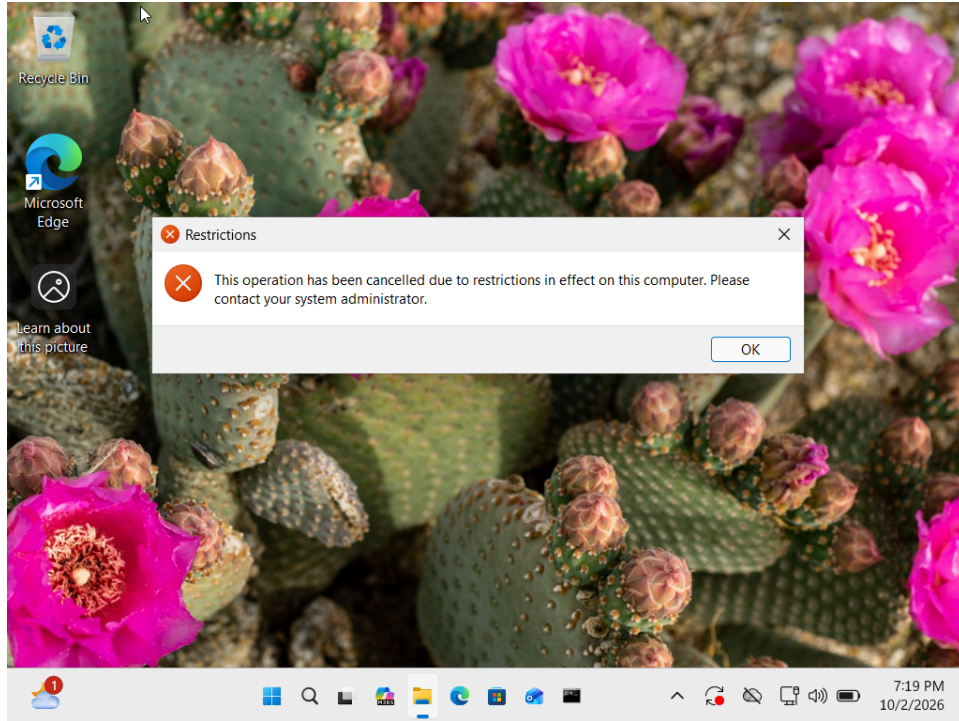


Screenshot 2 shows me accessing the “Shared Company Items” folder from the Network instead. Picture shows the MYSERVER domain controller on the network, along with the folders I have access to.

GPO 2: No Control Panel

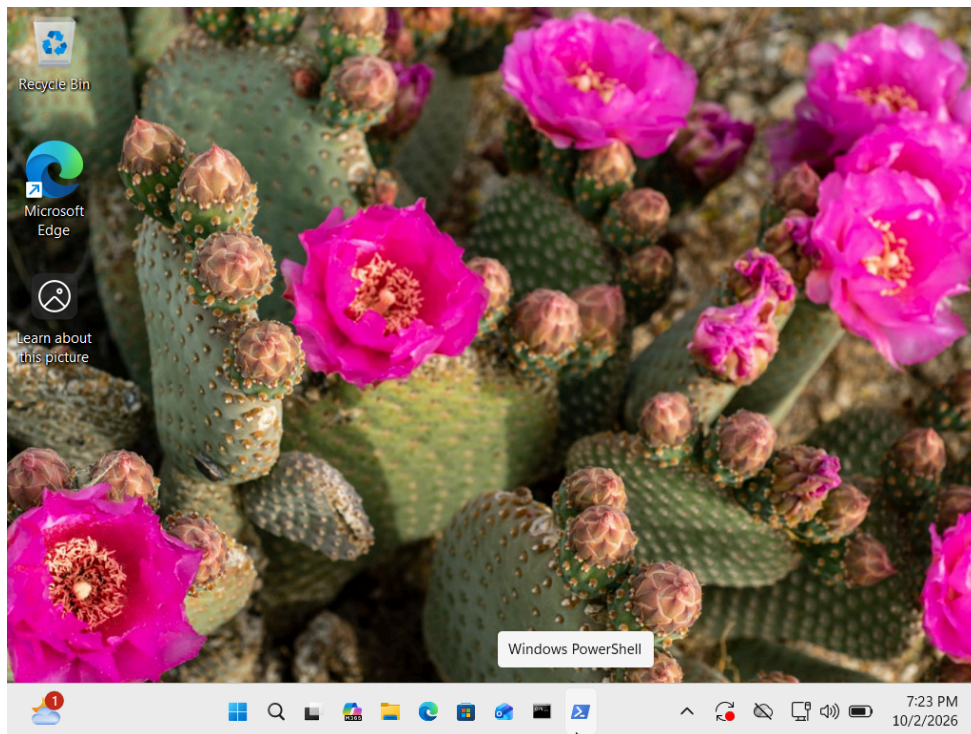


I clicked on the Windows logo, searched for the Control Panel and double clicked on the icon in an effort to open it.

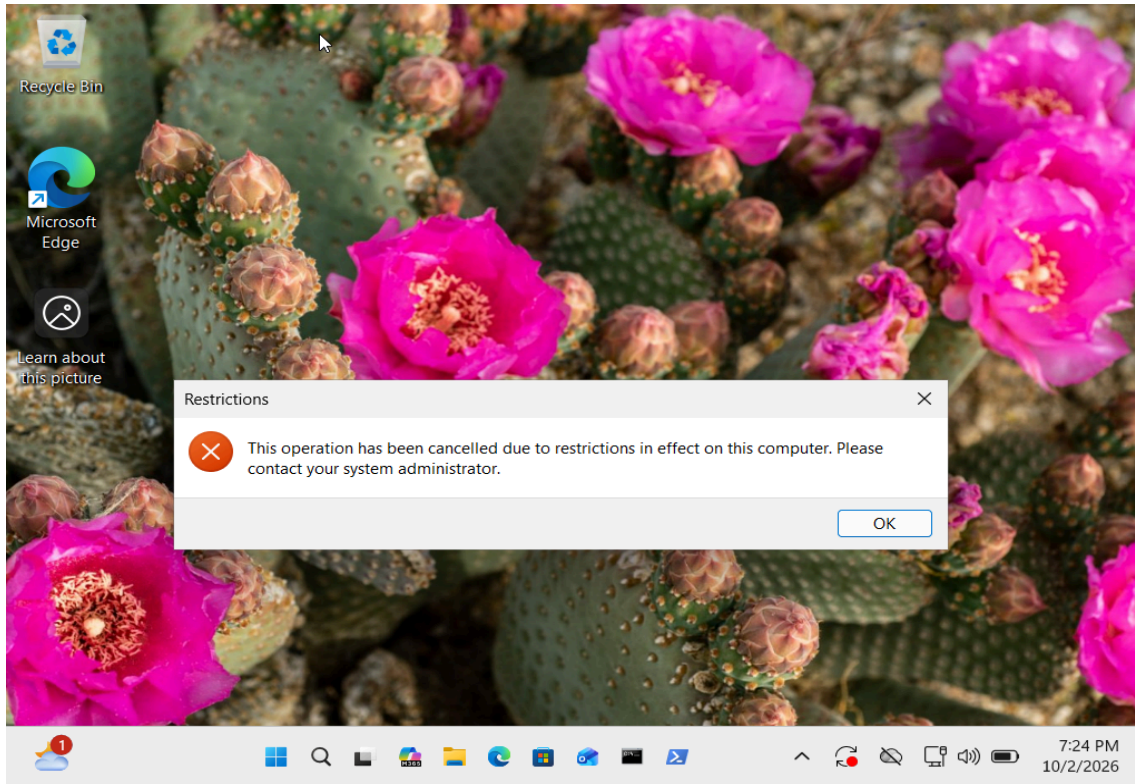


Error Message: “This operation has been cancelled due to restrictions in effect on this computer. Please contact your system administrator.”

GPO 3: No Powershell



Screenshot 1 shows me attempting to click on the Powershell application on the Windows taskbar.



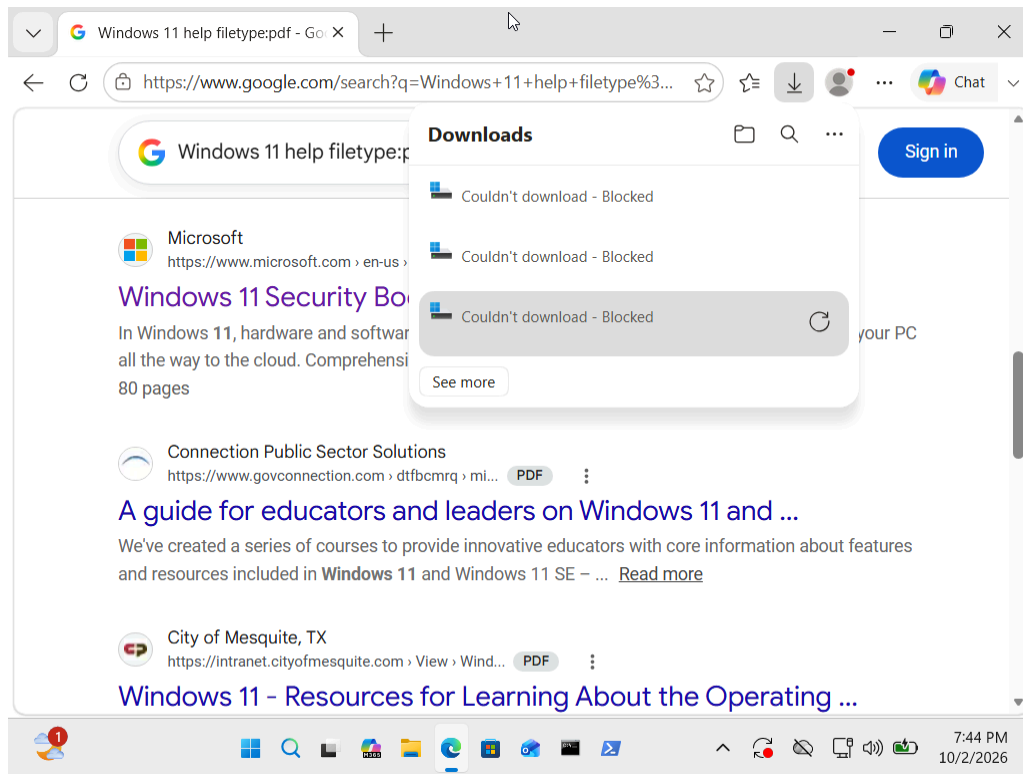
Same error message as above: “This operation has been cancelled due to restrictions in effect on this computer. Please contact your system administrator.”

Computer GPOs

GPO 4: Prohibit Downloads from Internet	Cannot download anything from Microsoft Edge
GPO 5: No USB	Cannot open USB on computer
GPO 6: Password Policy	Requires complex passwords (uppercase, lowercase, numbers, symbols), minimum lengths, maximum attempts before lockout, and no password-reuse

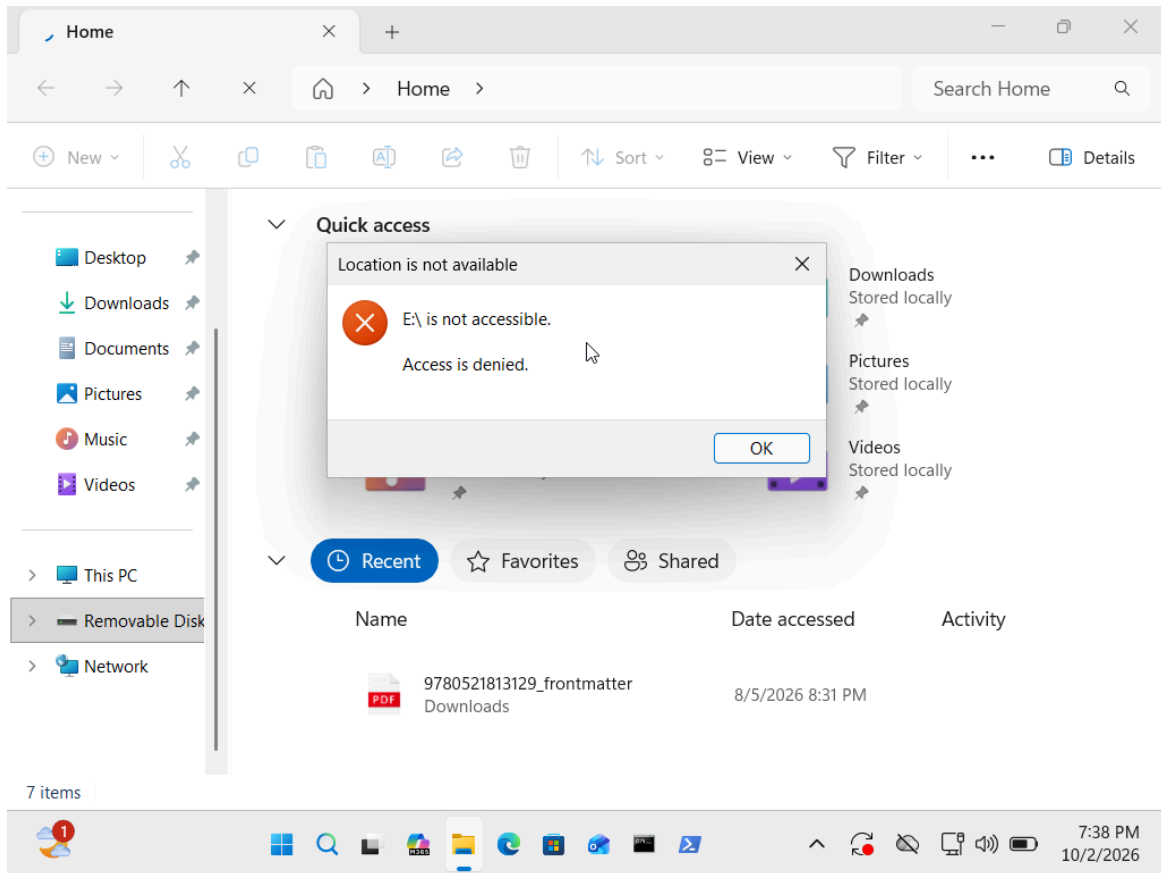
(with reference to the 24 most recent passwords).

GPO 4: Prohibit Downloads from Internet



I tried to download Microsoft's Windows 11 Security Book multiple times and was routinely blocked.

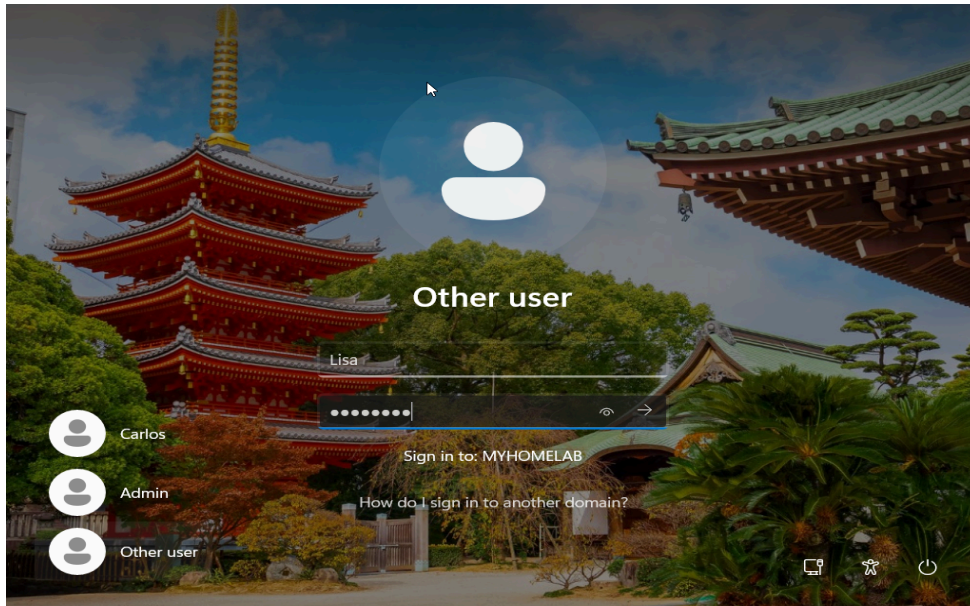
GPO 5: No USB



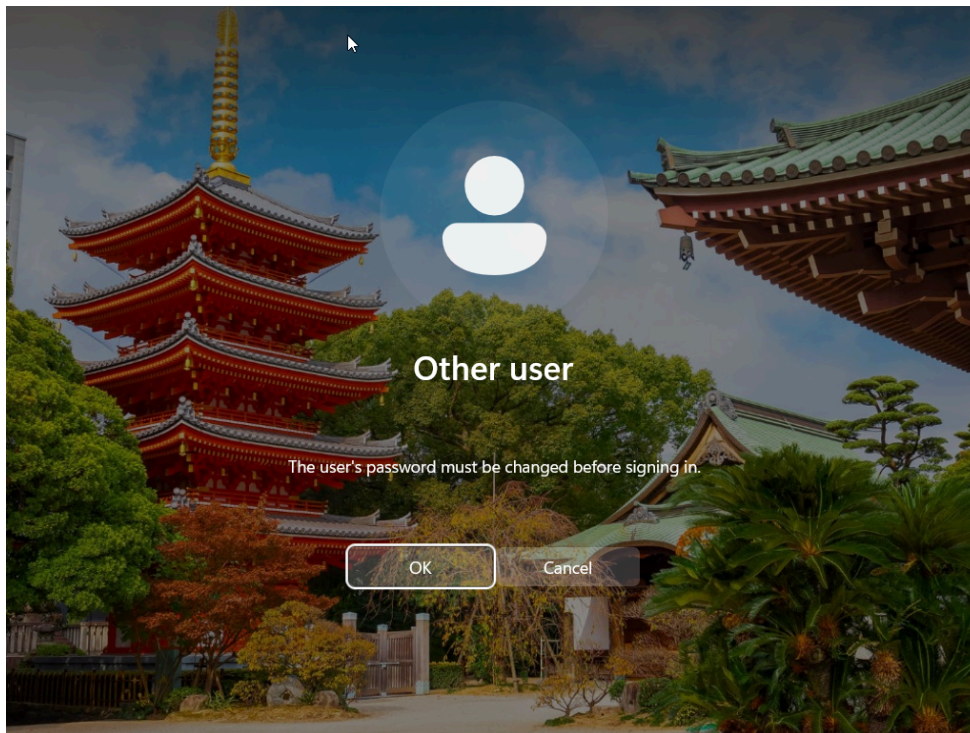
Attempting to open the “Removable Disk” produces an error message: “E:\ is not accessible. Access is denied.”

GPO 6: Password Policy

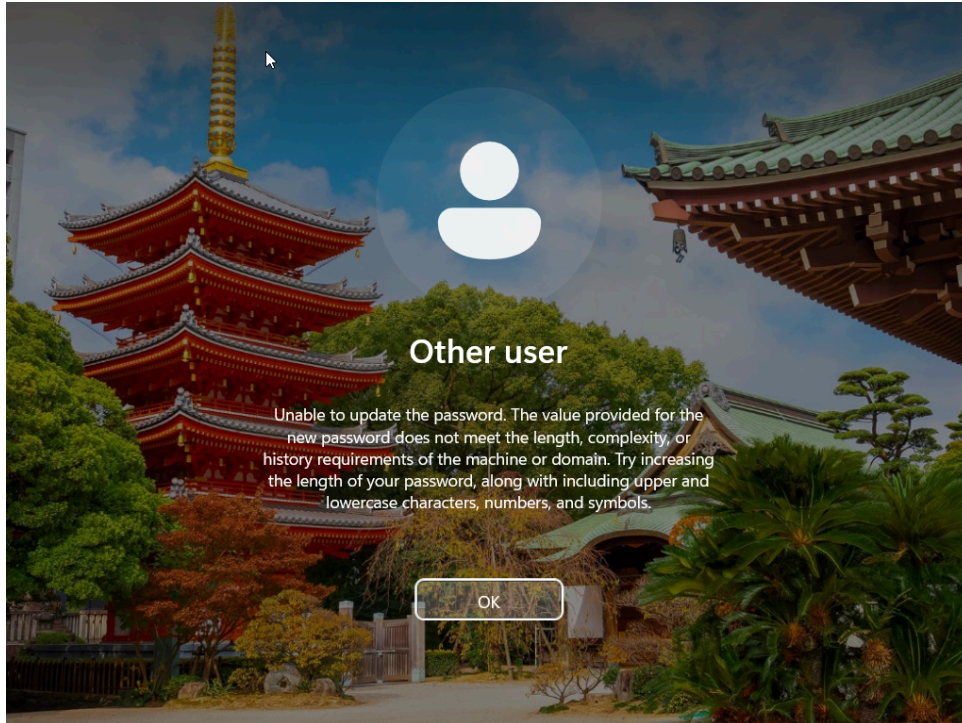
For this one, I decided to log out of Carlos’ account and demonstrate with another employee in my Active Directory, Lisa. After creating and enabling the password policy, I was immediately prompted to reset my password before moving forward. I attempted to implement the same password, and was blocked. Only after changing to an acceptable password that met all criteria was I able to log into Lisa’s account.



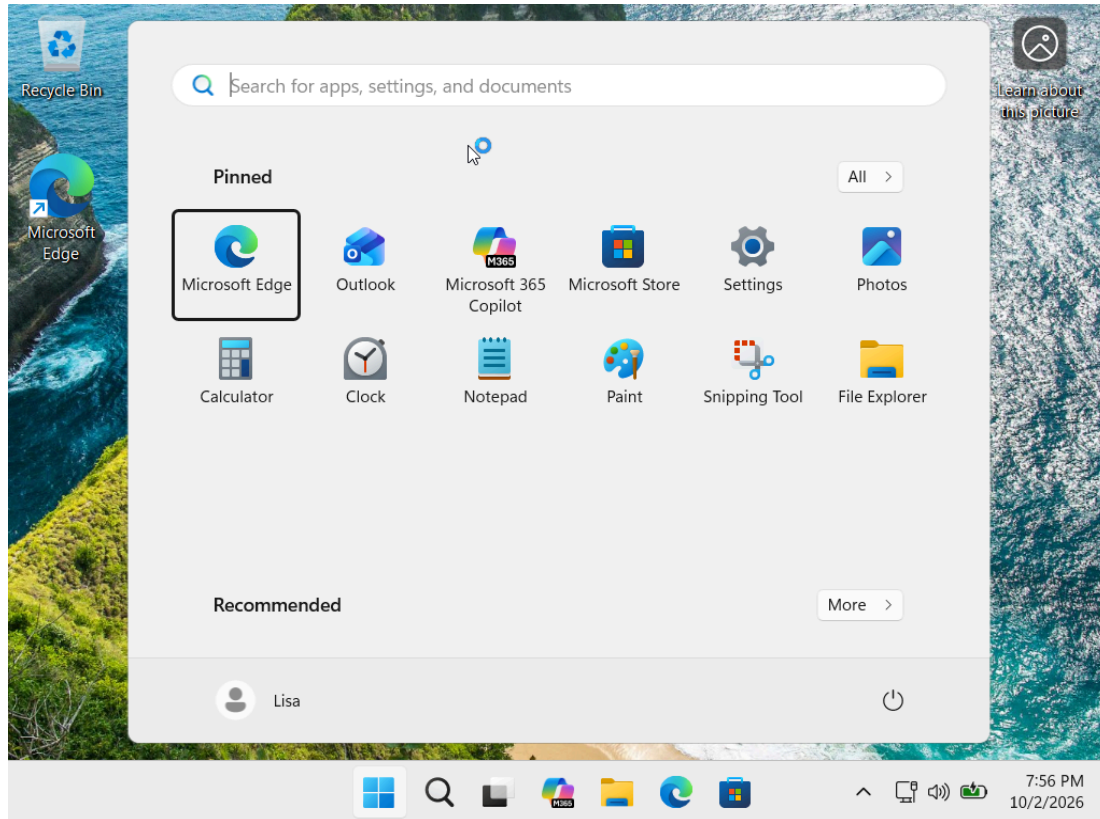
First attempt to log into Lisa's account.



Message: "The user's password must be changed before signing in."



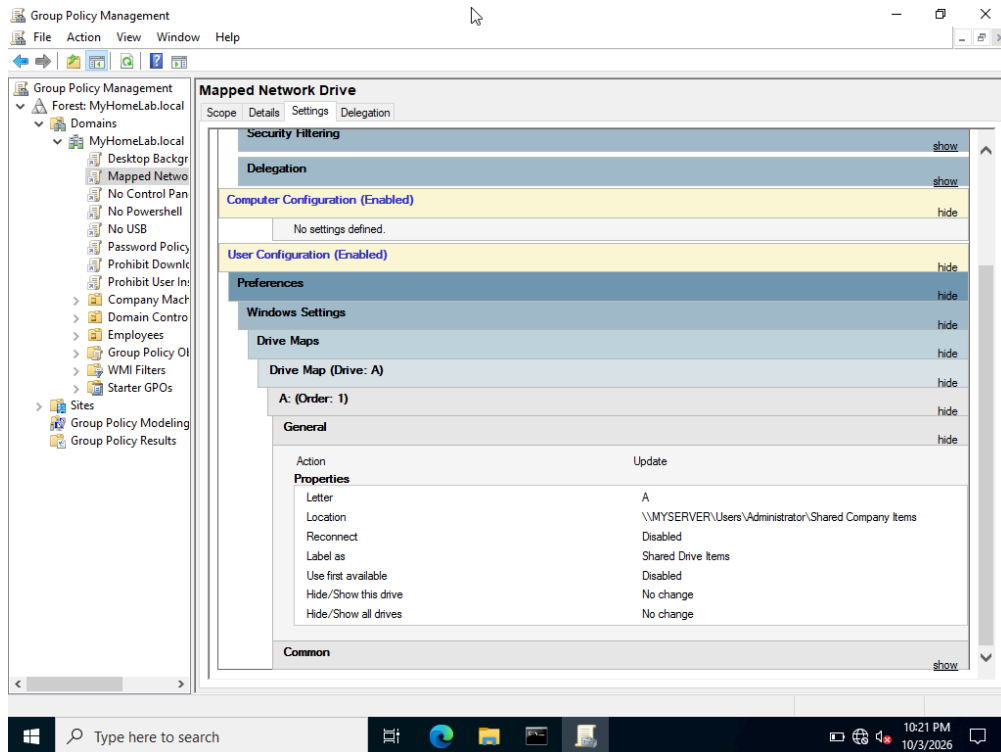
Error message: Unable to update the password. The value provided for the new password does not meet the length, complexity, or history requirements of the machine or domain. Try increasing the length of your password, along with including upper and lowercase characters, numbers, and symbols.



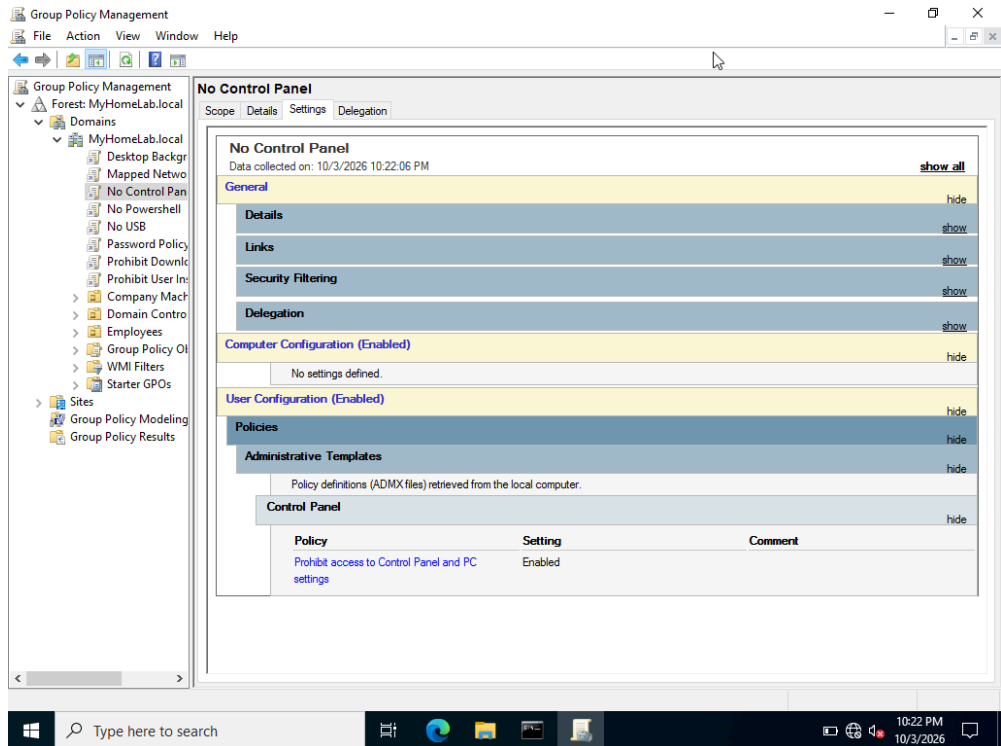
Successful login to Lisa's account!

GPO Settings on Domain Controller

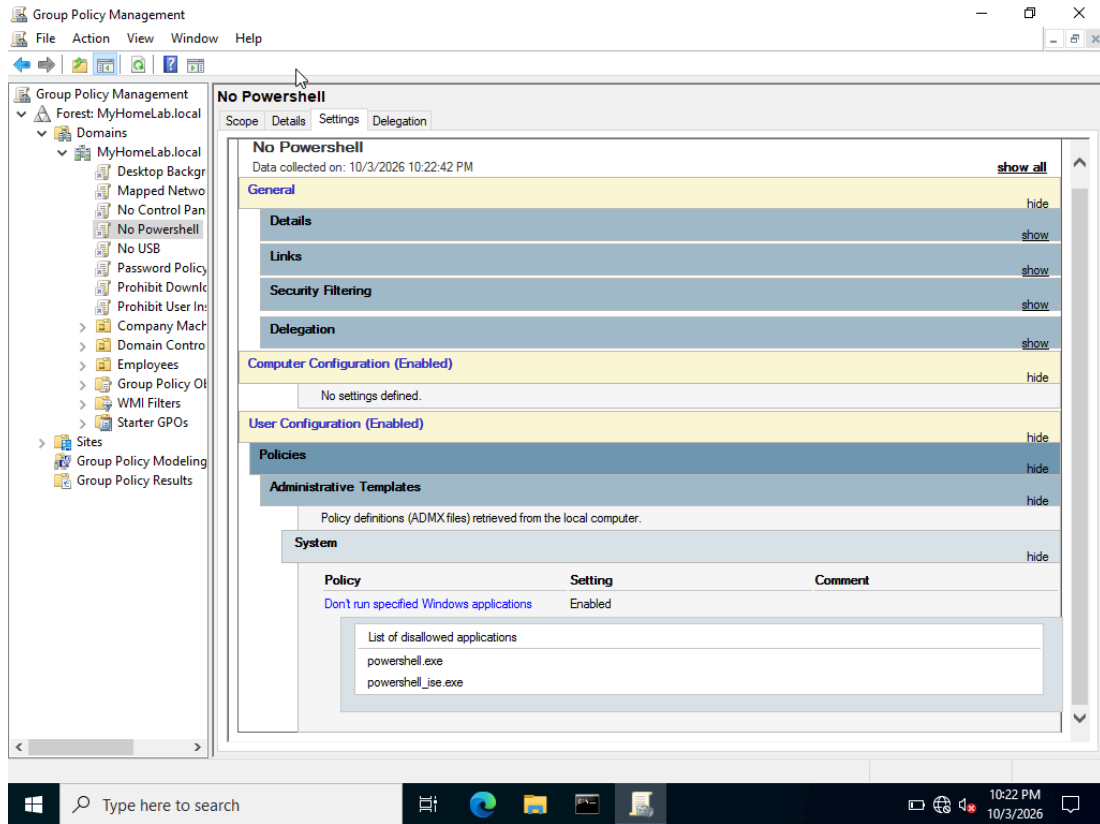
Mapped Network Drive



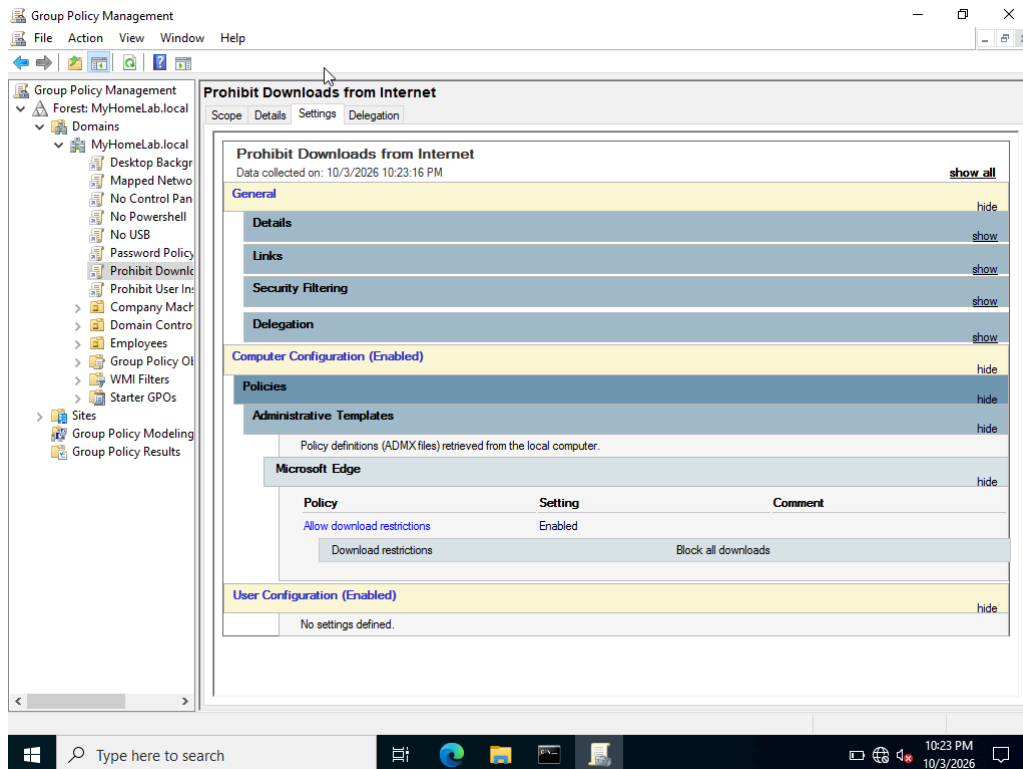
No Control Panel



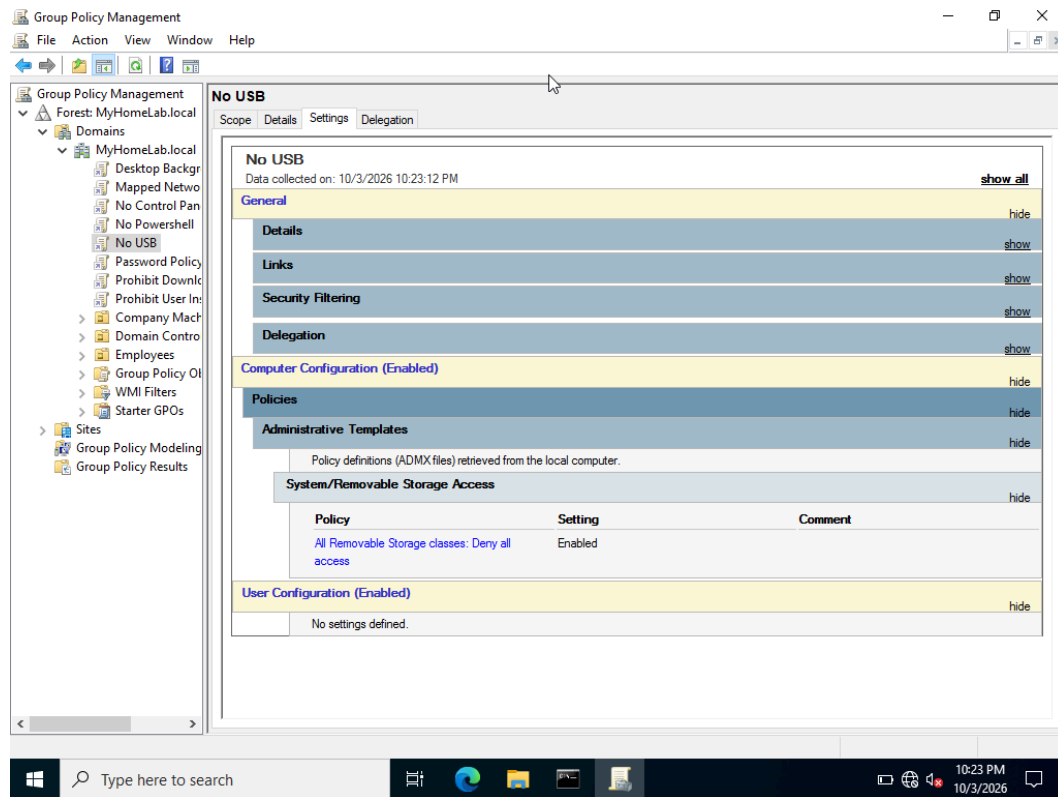
No Powershell



Prohibit Downloads from Internet



No USB



Password Policy

Group Policy Management

File Action View Window Help

Group Policy Management

- Forest: MyHomeLab.local
 - Domains
 - MyHomeLab.local
 - Desktop Backgr
 - Mapped Netwo
 - No Control Pan
 - No Powershell
 - No USB
 - Password Policy
 - Prohibit Downlo
 - Prohibit User In
 - Company Mach
 - Domain Contro
 - Employees
 - Group Policy Ol
 - WMI Filters
 - Starter GPOs
 - Sites
 - Group Policy Modeling
 - Group Policy Results

Password Policy

Scope Details Settings Delegation

Security Settings show

Delegation show

Computer Configuration (Enabled) hide

Policies

Windows Settings hide

Security Settings hide

Account Policies/Password Policy hide

Policy	Setting
Enforce password history	24 passwords remembered
Maximum password age	90 days
Minimum password age	30 days
Minimum password length	6 characters
Password must meet complexity requirements	Enabled
Store passwords using reversible encryption	Disabled

Account Policies/Account Lockout Policy hide

Policy	Setting
Account lockout duration	30 minutes
Account lockout threshold	3 invalid logon attempts
Reset account lockout counter after	30 minutes

Account Policies/Kerberos Policy show

Local Policies/Security Options show

10:25 PM 10/3/2026